

# Moller Polarimeter meeting, April 20, 2012

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## 1. Moller Polarimeter Status

1. Moller polarimeter measures beam polarization;
2. Q1 magnet field mapping is [done](#)
3. Moller quads shift is smaller but not zero.
  - PREX Quads shift GEANT vs. Measurement ( $E_b=1.06\text{GeV}$ ) ([plot](#))
  - g2p Quads shift GEANT vs. Measurement ( $E_b=2.25\text{GeV}$ ) ([plot](#))
  - g2p Quads shift GEANT vs. Measurement ( $E_b=1.71\text{GeV}$ ) ([plot](#))
4. Progress with old DAQ upgrade
5. FADC data triggers analysis still is not available
6. Discrepancy between old and new DAQs (0.1-0.5%) ([plot](#)) and ([plot](#))

## 2. 12 GeV Upgrade Status and Summer Plans

1. Design
  1. New detector shielding box - design is finished
  2. Moller beam line - design is finished
  3. Dipole shielding insertion - design is finished
  4. New Moller quadrupole magnet - design is finished
  5. Bench for Q1 and Q2 magnets - design is finished
  6. Girder (corrector\_quad\_BPM) - design is finished
2. Summer Installation
  1. Turn off BOPs, Moller crates, unplug Moller target motors (will ask Jack)
  2. All Moller magnets water cooling test (at high field)
  3. Do not install Moller detector into the new shielding box
  4. New quad field mapping is highly desired
  5. Any (many) pictures of the dipole and shielding insertion will be highly appreciated

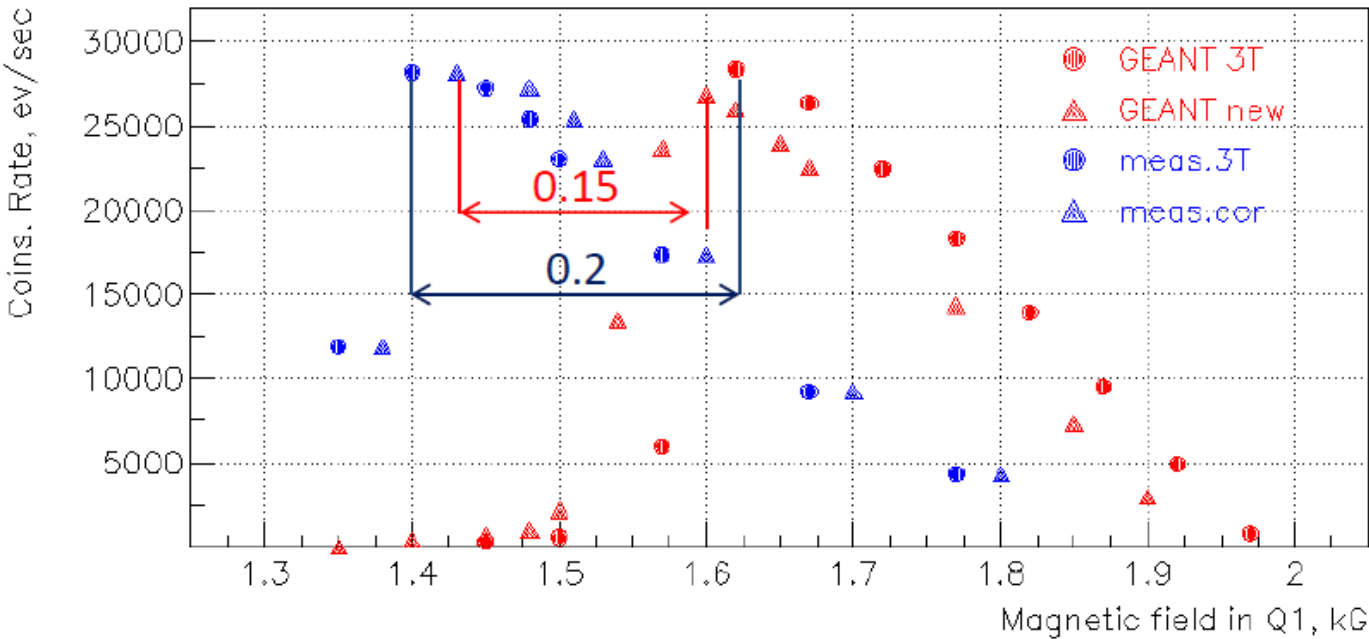
## 3. Future Plans

1. High Field Target:
    - New field mapper design
    - Installation
    - Field mapping + power supply commissioning
    - Target bench (target now is sitting +3mm upper of the beam line center)
    - Alignment
  2. The aperture detector upgrade + detector installation
  3. Moller settings function for MCC (0.8 - 11 GeV)
  4. Upgrade MCC Moller procedure
  5. New cryoline for the Moller target
  6. GEANT for the target alignment test with beam and new BPM
  7. Old DAQ upgrade
  8. FADC data triggers analysis
  9. GEANT vs. measurement quads shift discrepancy
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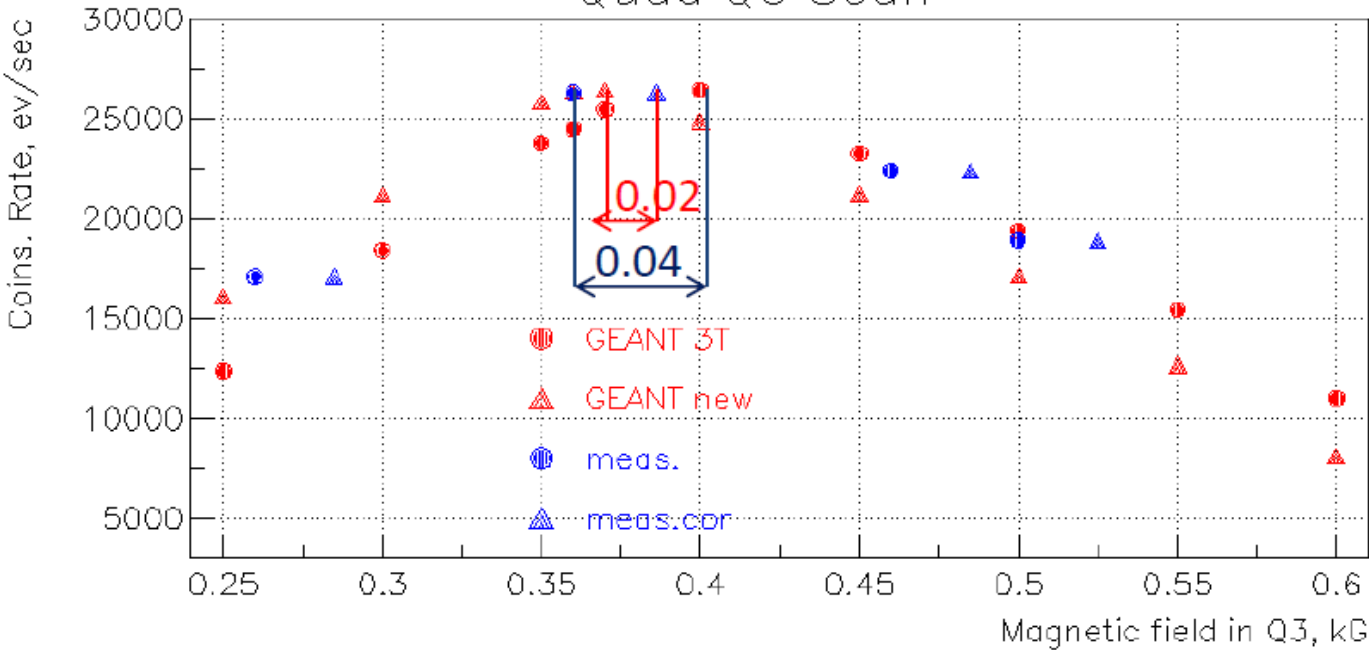
# Figures

PREX Quads shift GEANT vs. Measurement (Eb=1.06GeV)

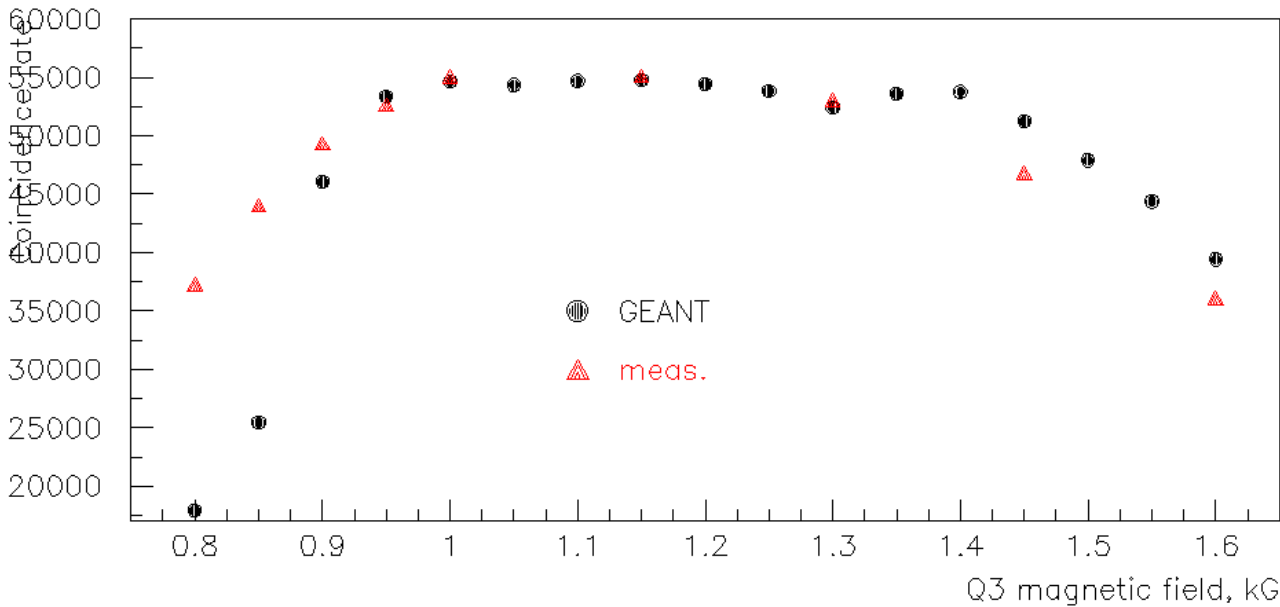
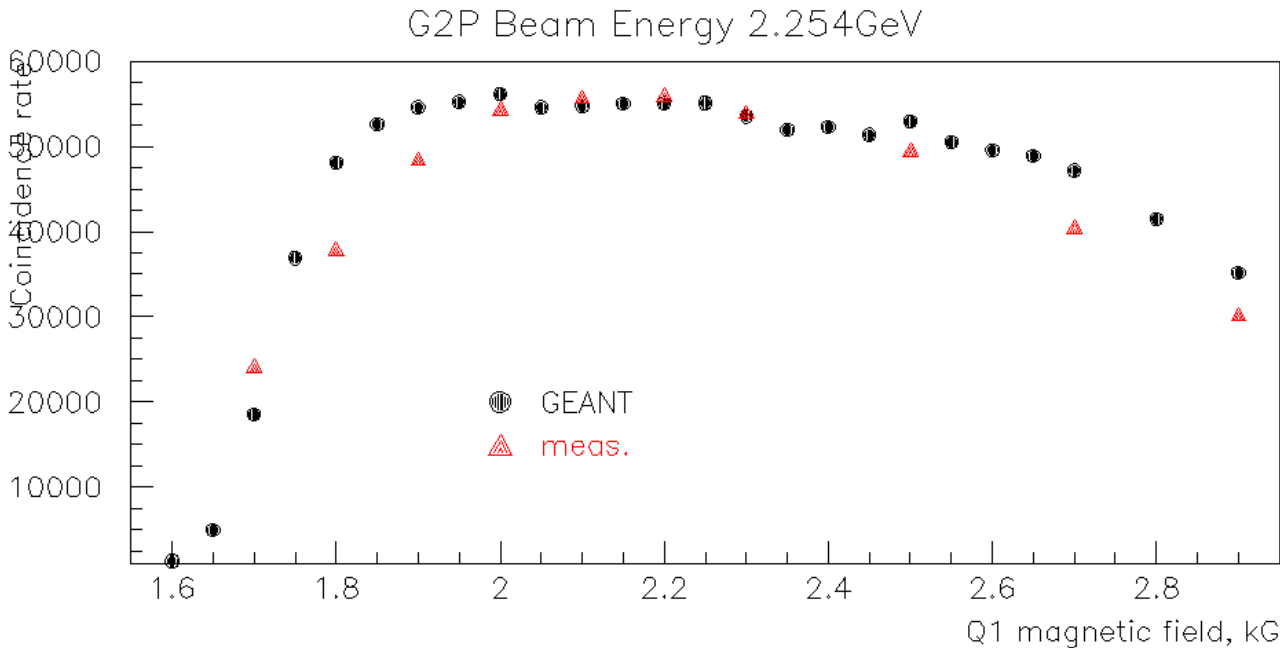
PREX Quad Q1 Scan.



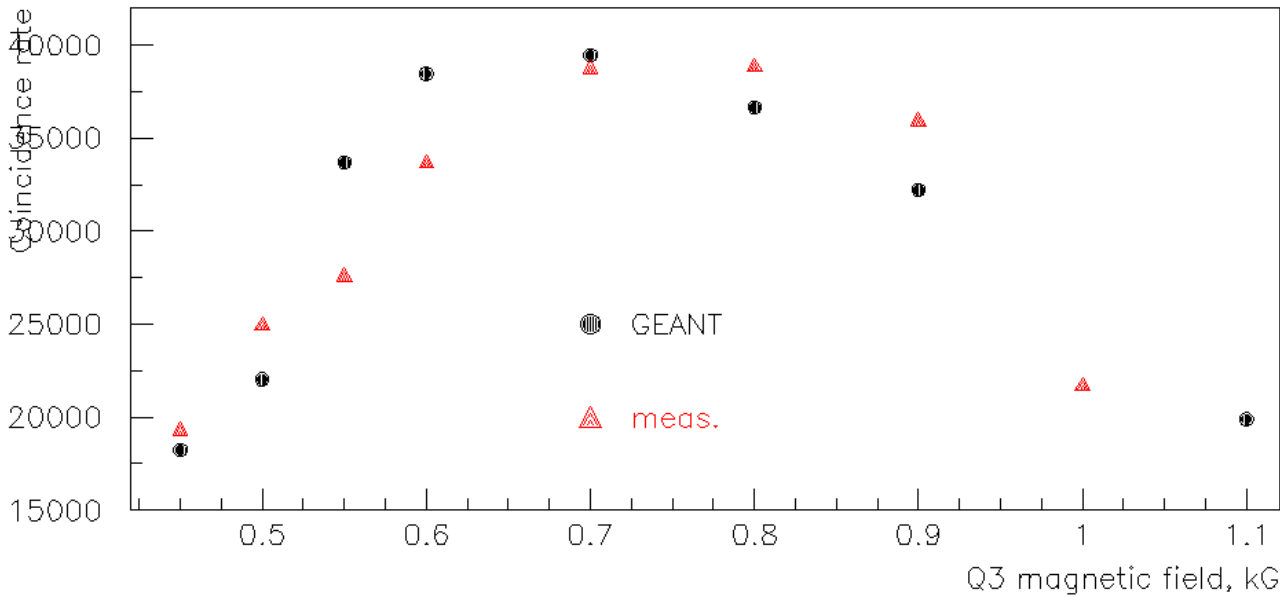
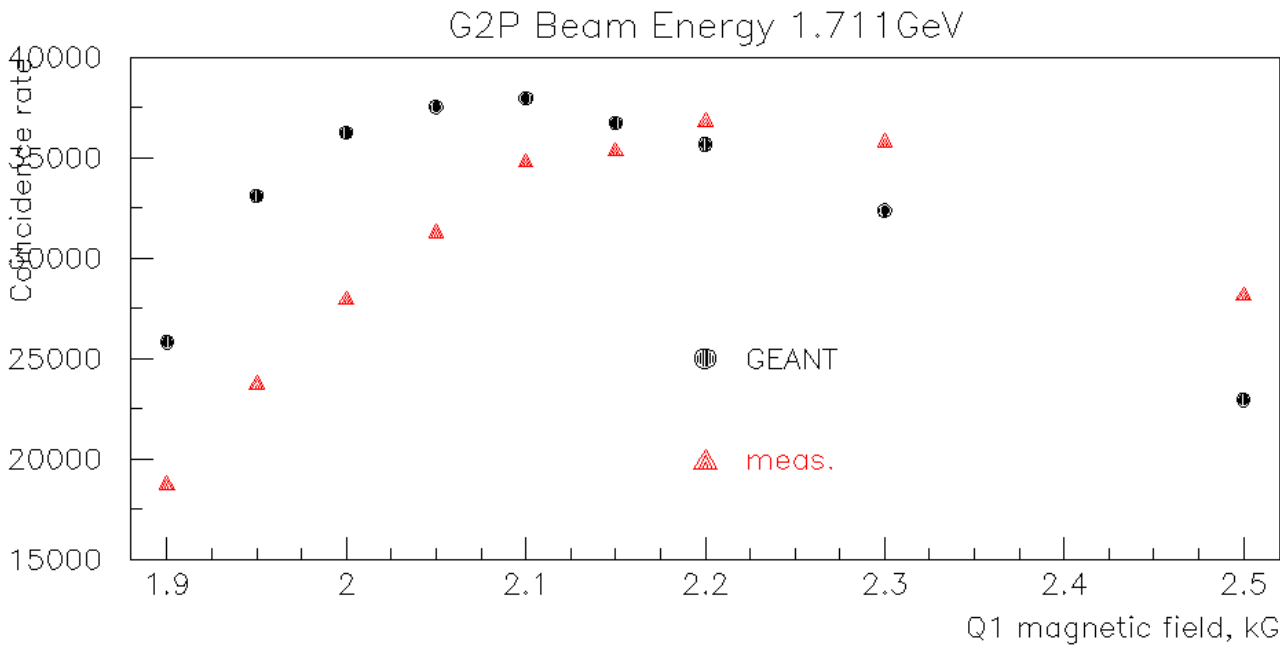
Quad Q3 Scan



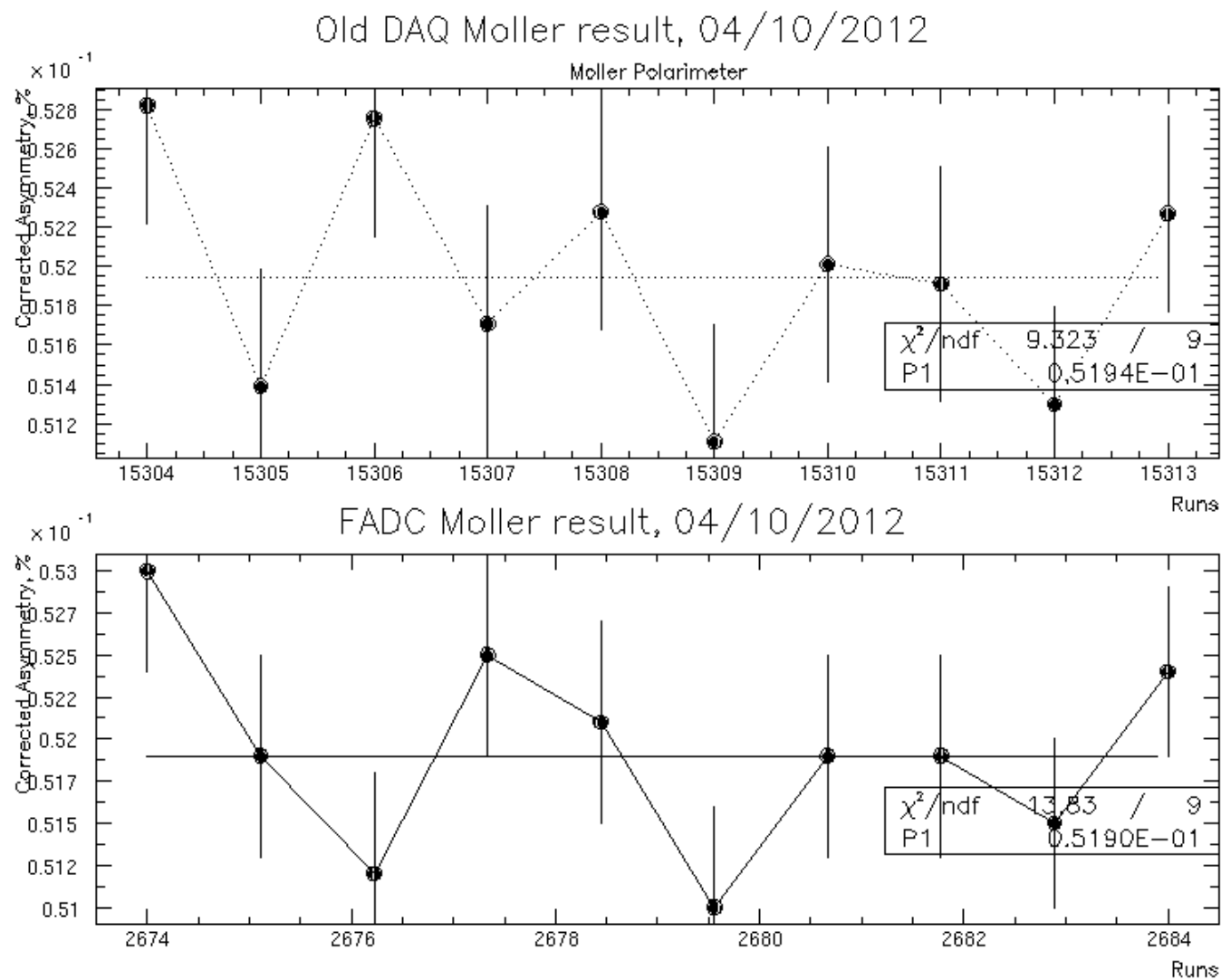
**g2p Quads shift GEANT vs. Measurement (Eb=2.25GeV)**



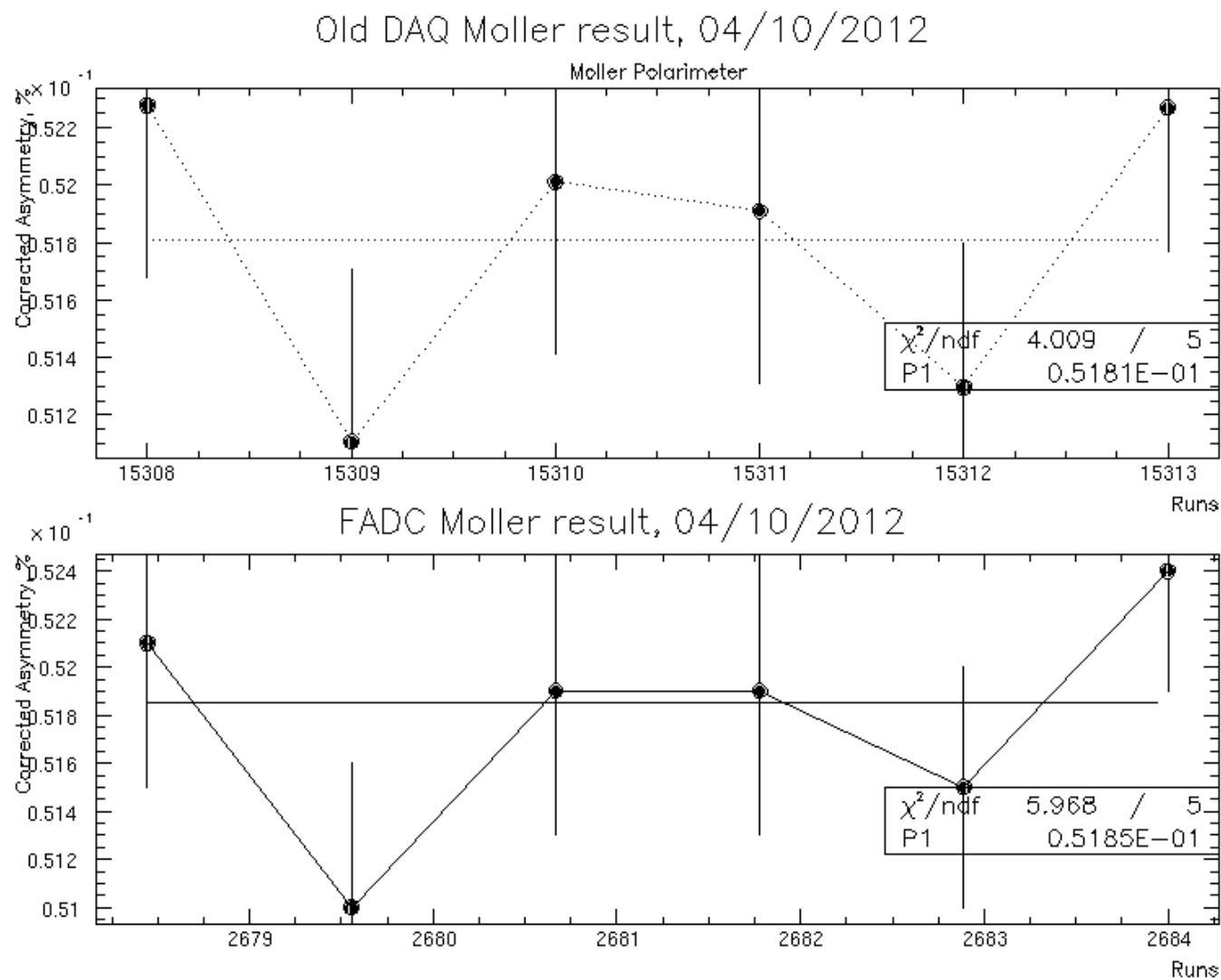
**g2p Quads shift GEANT vs. Measurement (Eb=1.71GeV)**



Old vs new DAQ discrepancy (0.1â€“0.5%)



Old vs new DAQ discrepancy (second plot)



|                             |              |              |                |               |              |
|-----------------------------|--------------|--------------|----------------|---------------|--------------|
| Møller notation             | Q1           | Q2           | Q3             | -             | Dipole       |
| Name                        | <i>PATSY</i> | <i>TESSA</i> | <i>FELICIA</i> | <i>JACKIE</i> | <i>LILLY</i> |
| Relevant information        |              |              |                |               |              |
| Element                     | quad         | quad         | quad           | quad          | dipole       |
| Bore, cm                    | 10.16        | 10.16        | 10.16          | 10.16         |              |
| Effective length, cm        | 45.72        | 35.66        | 35.66          | 45.72         | 164.27       |
| Maximum current, A          | 300          | 280          | 280            | 300           | 550          |
| Pole tip field at 100A, kGs | 2.029        | 2.371        | 2.420          | ≈2.03         | 2.727        |
| Pole tip field at 300A, kGs | 5.801        | 6.029        | 6.135          | 5.809         |              |
| Integrated GL at 100A, Gs   | 18308        | 17112        | 17524          | ≈18300        |              |
| Other technical information |              |              |                |               |              |
| Manufacturer                | MagnaTek     | Indust.Coil  | Indust.Coil    |               |              |
| Part number                 | LA-446205    | LA-446201    | LA-446200      |               |              |
| Yoke type                   | Solid        | Solid        | Solid          |               |              |
| Yoke material               | 1006         | 1006         | 1006           |               |              |
| Weight, kg                  | 450          | 500          | 500            |               |              |
| Color                       | Red          | Blue         | Blue           | Blue          |              |
| Number of coils             | 4            | 4            | 4              | 4             | 4            |
| Num. of turns per coil      | 42           | 50           | 50             | 42            | 32           |
| Resistance @20C°, ohm       | 0.1133       | 0.0767       | 0.0767         |               |              |
| Shunts and sec. coils       | none         | none         | none           | none          | none         |
| Voltage drop at max cur     | 10.6VDC      | 8.5VDC       | 8.5VDC         |               |              |
| Total water flow            | 1.5gpm       | 1.0gpm       | 1.0gpm         |               |              |
| Max. oper. pressure         | 35psi        | 35psi        | 35psi          |               |              |
| Cooling medium              | LCW          | LCW          | LCW            |               |              |

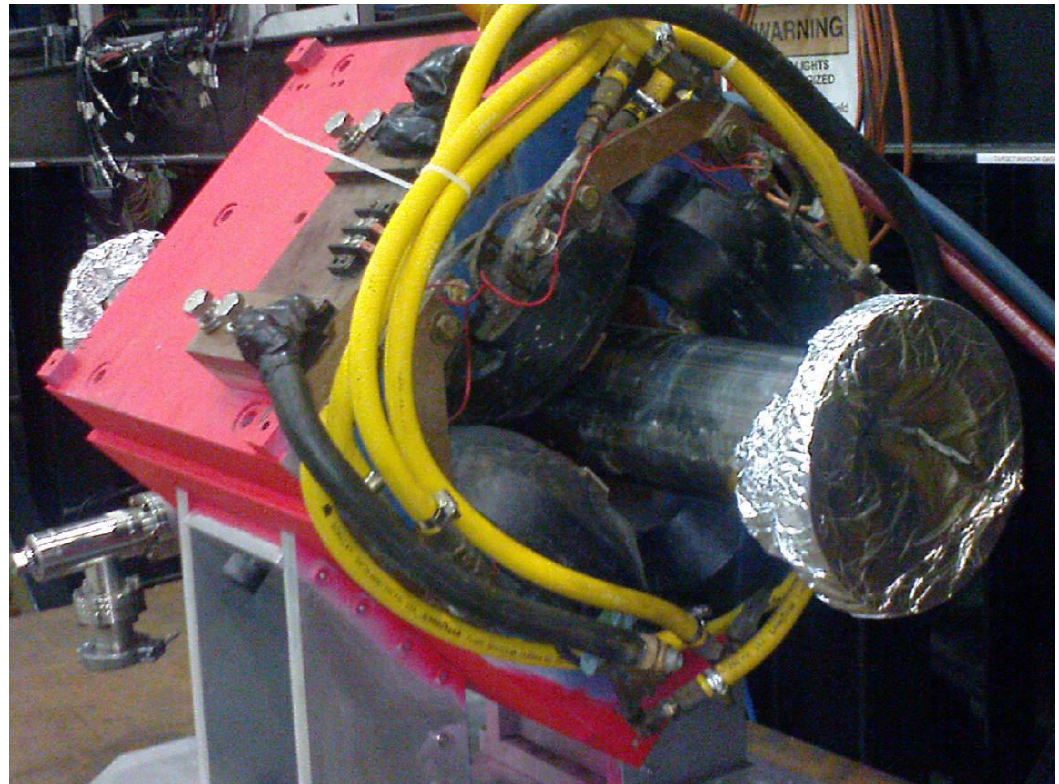
Table 1: Hall A Moller Quadrupole Parameters

| Quad 1 Radius= 4.96cm Length= 44.77cm |            |                           |           |          |          |            |
|---------------------------------------|------------|---------------------------|-----------|----------|----------|------------|
| Current                               | $B_{pole}$ | $\int G \cdot dl$ (Gauss) |           |          |          |            |
| (A)                                   | (Gauss)    | Quadrupole                | Sextupole | Octupole | Decapole | Dodecapole |
| 0.00                                  | 32.00      | 290.00                    | 0.32      | 0.17     | 0.03     | 1.51       |
| 49.94                                 | 1014.50    | 9143.00                   | 10.97     | 1.83     | 1.83     | 16.46      |
| 100.24                                | 1997.00    | 18070.00                  | 28.91     | 3.61     | 3.61     | 32.53      |
| 150.14                                | 2976.00    | 26910.00                  | 40.36     | 5.38     | 5.38     | 48.44      |
| 200.09                                | 3954.00    | 35710.00                  | 53.57     | 14.28    | 7.14     | 74.99      |
| 249.93                                | 4918.00    | 44350.00                  | 70.96     | 22.17    | 8.87     | 115.31     |
| 299.15                                | 5801.50    | 52040.00                  | 98.88     | 26.02    | 10.41    | 161.32     |
| Quad 2 Radius= 5.00cm Length= 36.74cm |            |                           |           |          |          |            |
| Current                               | $B_{pole}$ | $\int G \cdot dl$ (Gauss) |           |          |          |            |
| (A)                                   | (Gauss)    | Quadrupole                | Sextupole | Octupole | Decapole | Dodecapole |
| 0.00                                  | 31.00      | 228.00                    | 1.35      | 0.09     | 0.16     | 1.48       |
| 50.56                                 | 1187.00    | 8789.00                   | 25.49     | 0.88     | 1.76     | 23.73      |
| 100.26                                | 2329.50    | 17140.00                  | 63.42     | 5.14     | 1.71     | 47.99      |
| 150.08                                | 3450.00    | 25340.00                  | 101.36    | 5.07     | 2.53     | 68.42      |
| 200.04                                | 4530.00    | 33250.00                  | 139.65    | 6.65     | 6.65     | 93.10      |
| 250.20                                | 5418.50    | 39750.00                  | 155.03    | 7.95     | 7.95     | 123.22     |
| 299.04                                | 6029.50    | 43980.00                  | 153.93    | 8.80     | 8.80     | 145.13     |
| Quad 3 Radius= 5.00cm Length= 36.50cm |            |                           |           |          |          |            |
| Current                               | $B_{pole}$ | $\int G \cdot dl$ (Gauss) |           |          |          |            |
| (A)                                   | (Gauss)    | Quadrupole                | Sextupole | Octupole | Decapole | Dodecapole |
| 0.00                                  | 30.00      | 229.00                    | 1.92      | 0.25     | 0.25     | 1.56       |
| 50.11                                 | 1208.00    | 8783.00                   | 21.96     | 0.88     | 0.88     | 22.84      |
| 100.28                                | 2379.00    | 17290.00                  | 43.23     | 1.73     | 1.73     | 43.23      |
| 150.07                                | 3528.50    | 25570.00                  | 63.92     | 2.56     | 2.56     | 66.48      |
| 199.95                                | 4634.00    | 33540.00                  | 83.85     | 6.71     | 3.35     | 97.27      |
| 249.99                                | 5533.50    | 40030.00                  | 100.08    | 8.01     | 8.01     | 128.10     |
| 299.03                                | 6135.00    | 44240.00                  | 110.60    | 8.85     | 4.42     | 154.84     |

# Q1=PATSY=MQM1H02 Field Mapping Results

Was done:

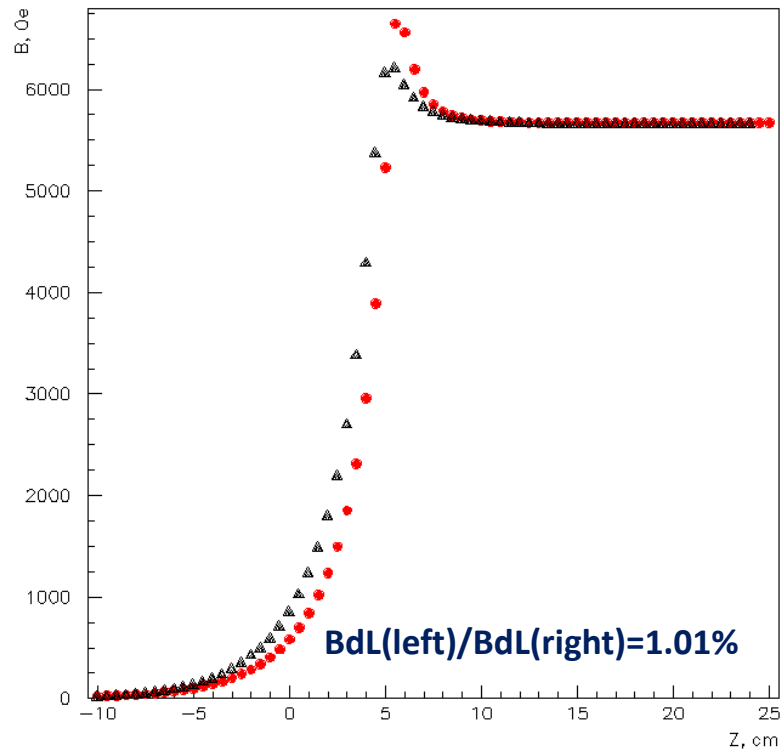
- GL measurement (string  $R=4.85\text{cm}$ )
- B measurement (Hall probe  $R\sim 4.85\text{cm}$ ) along the quad with fringe fields
- B measurement near the pole on the beam line (Hall probe)



## Results

| Set     | Readback   | GL, G    | Field, G | Field, G | HPr   |
|---------|------------|----------|----------|----------|-------|
| Sur.(A) | Current(A) | measur   | R=4.85cm | R=5.08cm | G     |
| -300.   | -298.92    | -52279.4 | -5657.64 | -5938.18 | -5560 |
| -250.   | -249.36    | -43945.1 | -4755.71 | -4991.53 | -4669 |
| -200.   | -198.01    | -35298.7 | -3820.00 | -4009.42 | -3750 |
| -150.   | -149.68    | -26480.6 | -2865.71 | -3007.81 | -2811 |
| -100.   | -100.11    | -17643.0 | -1909.31 | -2003.99 | -1873 |
| -50.    | -50.33     | -8755.3  | -947.49  | -994.48  | -930  |
| -10.    | -10.49     | -1648.3  | -178.38  | -187.22  | -175  |
| 0.      | 0.0        | 235.7    | 25.51    | 26.78    | 24    |
| 10.     | 10.47      | 2119.8   | 229.40   | 240.78   | 221   |
| 50.     | 50.25      | 9237.0   | 999.62   | 1049.19  | 979   |
| 100.    | 100.02     | 18137.9  | 1962.87  | 2060.20  | 1929  |
| 150.    | 149.70     | 26986.4  | 2920.45  | 3065.26  | 2877  |
| 200.    | 199.43     | 35810.7  | 3875.41  | 4067.58  | 3828  |
| 250.    | 249.05     | 44466.6  | 4812.15  | 5050.76  | 4760  |
| 300.    | 298.92     | 52321.4  | 5667.07  | 5942.95  | 5604  |

|                |        |           |         |          |                        |      |
|----------------|--------|-----------|---------|----------|------------------------|------|
| Transducer (V) | I (A)  | BL (G-cm) | B (G)   | Leff, cm | Chamber Dimension (in) |      |
| 5.98           | 298.99 | 253514.7  | -5668.4 | 44.77    | OD                     | 4    |
|                |        |           |         |          | ID                     | 3.87 |



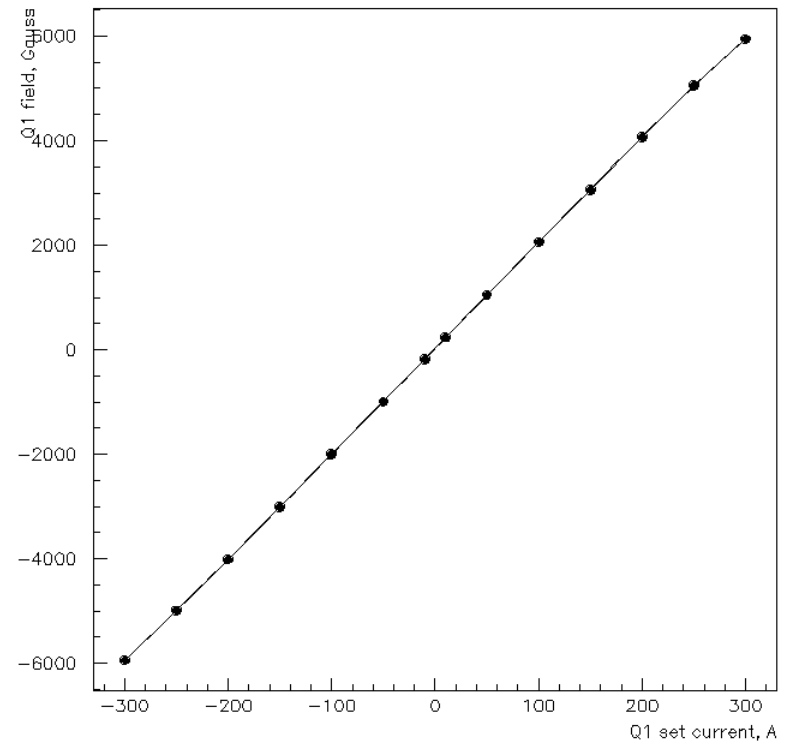
**Q1:**

$L_{eff}(old) \ 45.05cm \rightarrow 44.77cm$

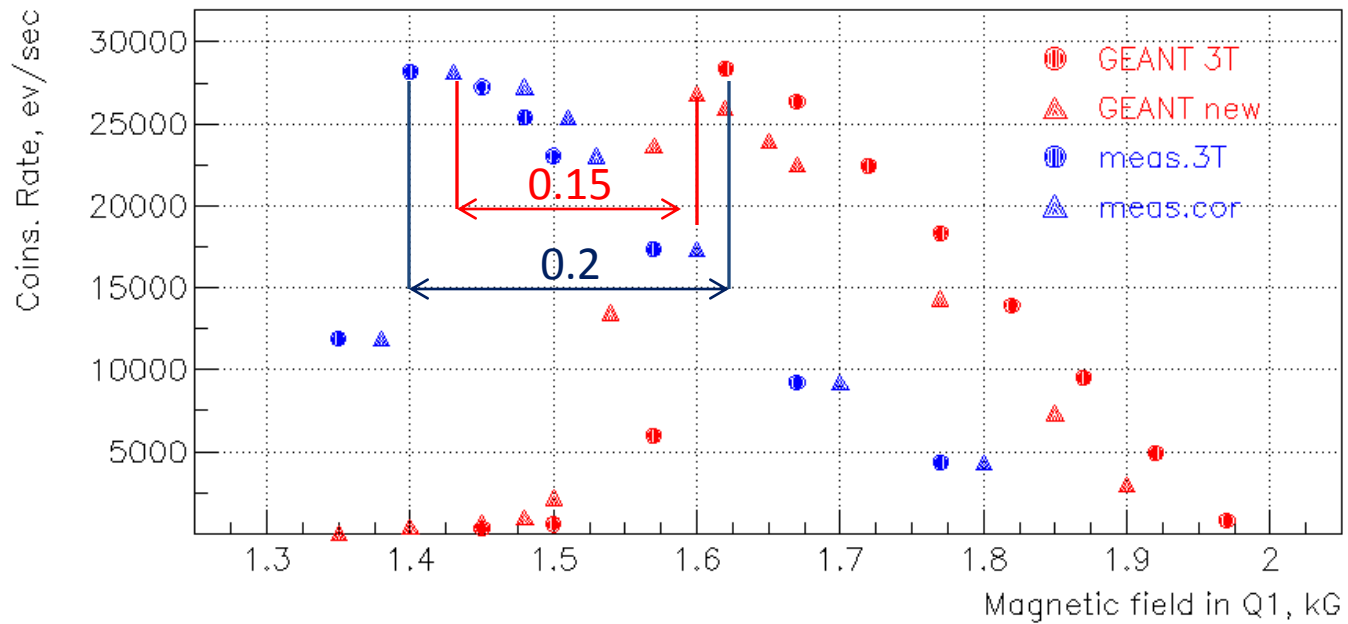
GEANT field  $R=5.08 \rightarrow$  acceler.  $R=4.96cm$

Coef.  $B \rightarrow GL \rightarrow A$  constant  $\rightarrow$  fit

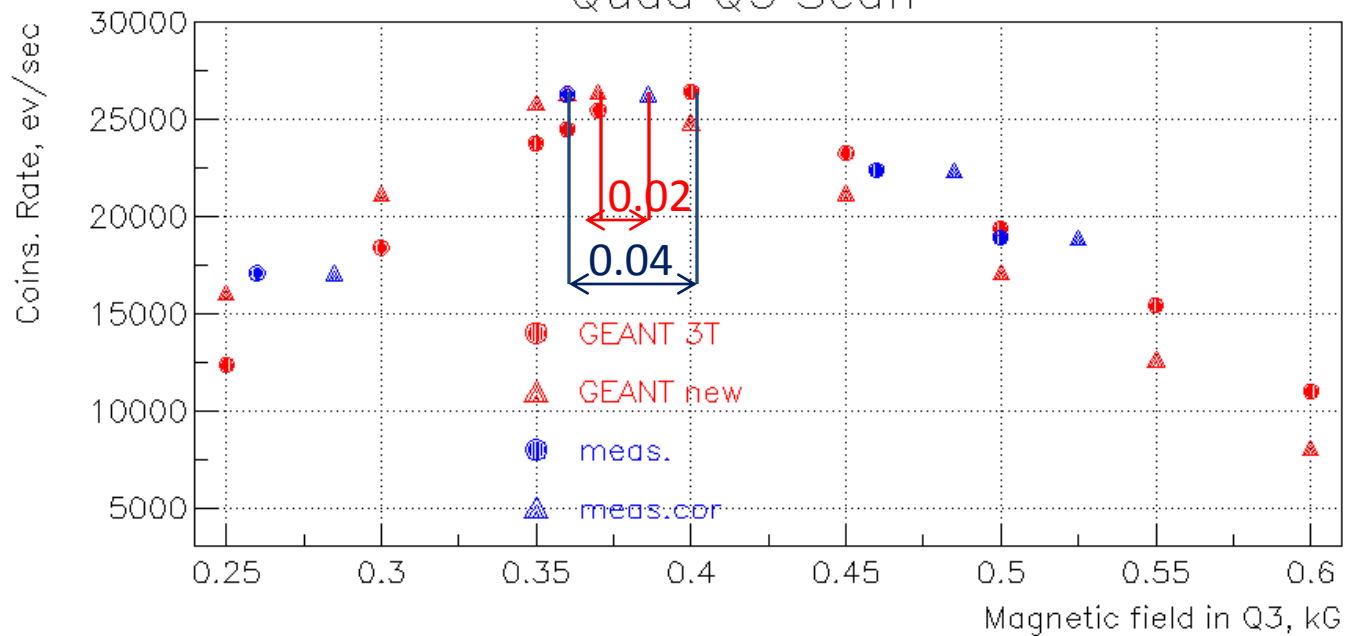
$0A=0Gauss \rightarrow 0A=26Gauss$



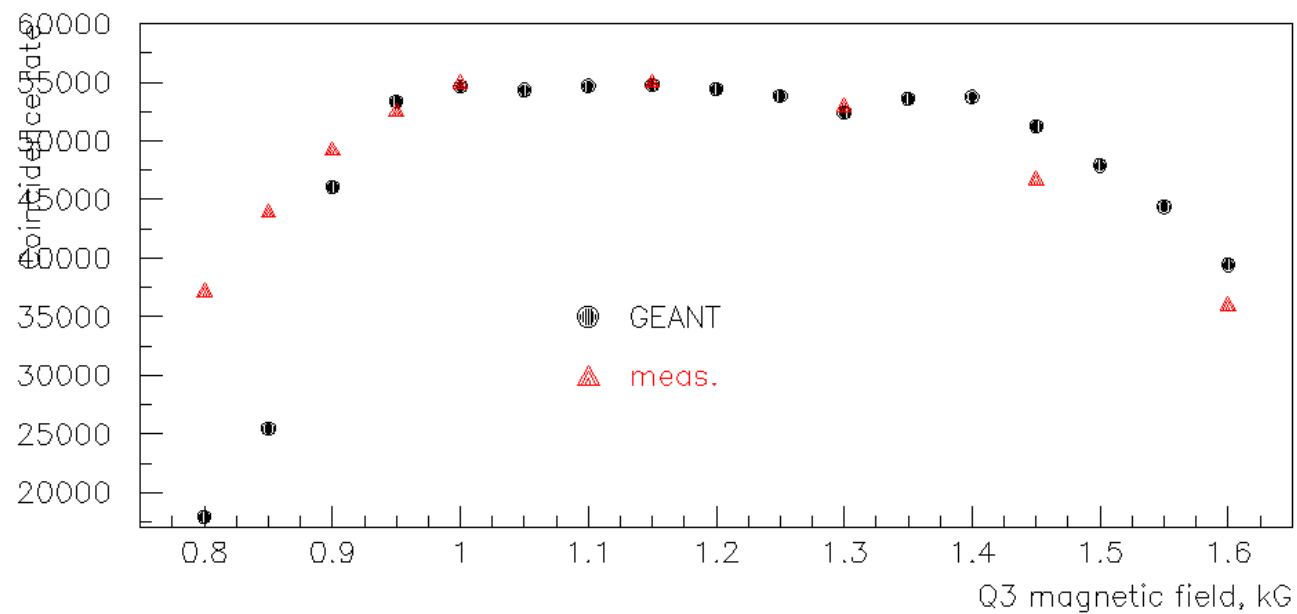
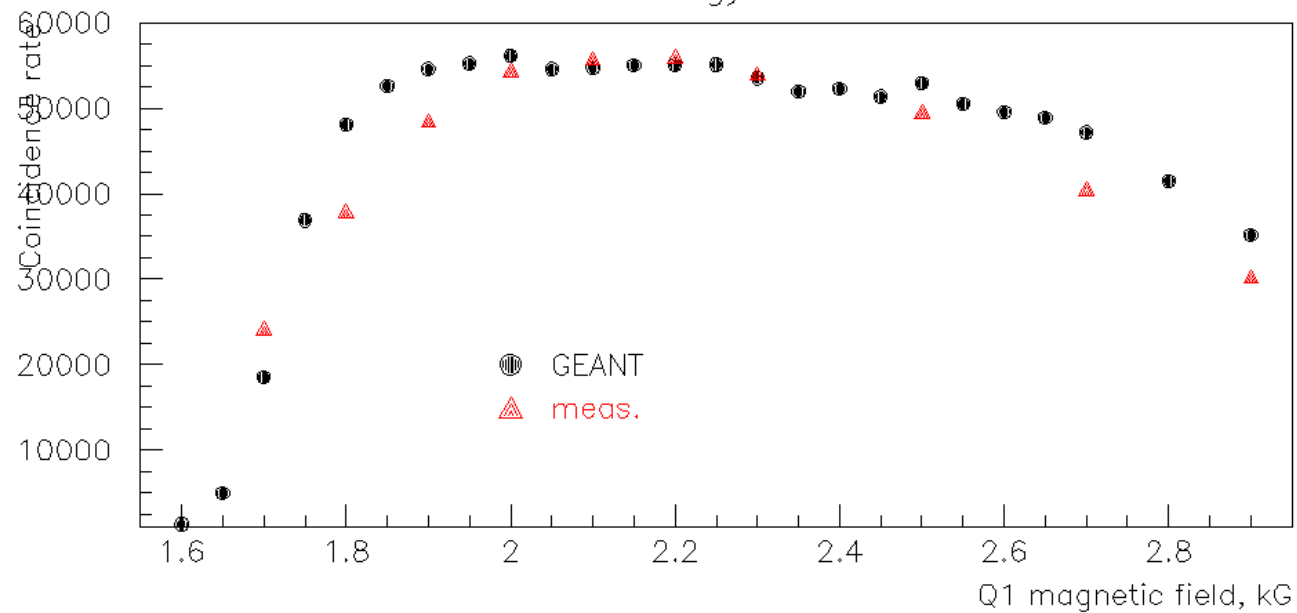
# PREX Quad Q1 Scan.



# Quad Q3 Scan



G2P Beam Energy 2.254GeV



G2P Beam Energy 1.711GeV

